

## CENG381 Stochastic Processes

### Renewal-Reward Processes, Residual Life, and Inspection Paradox - Question Set 1

---

Notation used in this question set:

|                                                     |                                                                                                                                                                                                                                                |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reward <math>R_n</math>:</b>                     | a random variable earned at the $n$ -th renewal. $\{R_n\}$ are i.i.d. with mean $E[R]$ . $R_n$ may be a cost, profit, or duration -- any quantity that resets at each renewal.                                                                 |
| <b><math>C(t)</math>:</b>                           | total cumulative reward earned by time $t$ : $C(t) = R_1 + R_2 + \dots + R_{N(t)}$ .                                                                                                                                                           |
| <b>Renewal-Reward Theorem:</b>                      | $C(t) / t \rightarrow E[R] / E[X]$ as $t \rightarrow \infty$ . Long-run reward rate = (mean reward per cycle) / (mean cycle length).                                                                                                           |
| <b><math>A(t)</math> = age (backward):</b>          | time elapsed since the LAST renewal before time $t$ : $A(t) = t - S_{N(t)}$ . Also called the current age or backward recurrence time.                                                                                                         |
| <b><math>B(t)</math> = residual life (forward):</b> | time remaining until the NEXT renewal after time $t$ : $B(t) = S_{N(t)+1} - t$ . Also called excess life or forward recurrence time.                                                                                                           |
| <b>Inspection Paradox:</b>                          | The interval $[S_{N(t)}, S_{N(t)+1}]$ that CONTAINS time $t$ tends to be LONGER than a typical inter-renewal interval. Specifically, its expected length is $E[X^2] / E[X] \geq E[X]$ (by Jensen's inequality, since $E[X^2] \geq (E[X])^2$ ). |

*For a stationary renewal process:  $E[B(t)] = E[A(t)] = E[X^2] / (2 * E[X])$ . Time-average of  $A(t)$  (over all  $t$ ) also equals  $E[X^2] / (2 * E[X])$  by the Renewal-Reward Theorem with reward = current age at renewal.*

**Q1 (30 points). A machine alternates between working (ON) and being repaired (OFF). ON periods have duration Exponential( $\mu_{on}$ ) with mean  $1/\mu_{on} = 4$  hours. OFF (repair) periods have duration Exponential( $\mu_{off}$ ) with mean  $1/\mu_{off} = 1$  hour. Cycles are i.i.d. A cycle = one ON period + one OFF period.**

- Define the renewal points as the moments when the machine starts a new ON period. What is the mean cycle length  $E[X]$  where  $X = \text{ON} + \text{OFF}$ ?
  - Define the reward  $R_n$  for cycle  $n$  as the length of the ON period in cycle  $n$ . Apply the Renewal-Reward Theorem to find the long-run fraction of time the machine is ON. Verify this matches intuition.
  - What is the long-run fraction of time the machine is OFF (being repaired)? Use the Renewal-Reward Theorem with reward = OFF duration for each cycle. Check that (fraction ON) + (fraction OFF) = 1.
-

## CENG381 Stochastic Processes

### Renewal-Reward Processes, Residual Life, and Inspection Paradox - Question Set 1

---

**Q2 (35 points).** Customers arrive at a shop according to a renewal process with inter-arrival times  $X_n \sim \text{Uniform}[0, 4]$  minutes (so  $E[X] = 2$  minutes,  $E[X^2] = \text{integral from 0 to 4 of } x^2 \cdot (1/4) dx = 16/3$ ).

- a) Compute  $E[X]$  and  $E[X^2]$  for  $X \sim \text{Uniform}[0, 4]$ . Recall:  $E[X^2] = \text{integral from 0 to 4 of } x^2 \cdot (1/4) dx$ .
  - b) You arrive at the shop at a large time  $t$  (so the process is approximately in stationarity). What is the expected residual waiting time  $B(t)$  until the next customer arrives after you? Use  $E[B] = E[X^2] / (2 \cdot E[X])$ .
  - c) What is the expected TOTAL length of the inter-arrival interval that contains time  $t$ ? (i.e., the interval from the last arrival before  $t$  to the next arrival after  $t$ .) Compare this to  $E[X]$  and explain the Inspection Paradox: why is the interval you land in longer on average than a typical interval?
- 

**Q3 (35 points).** Bus inter-arrival times are i.i.d. with the following distribution:

$P(X = 2) = 0.6$  (short gap, 2 minutes)

$P(X = 8) = 0.4$  (long gap, 8 minutes)

You arrive at the bus stop at a uniformly random time (large  $t$ , stationary).

- a) Compute  $E[X]$  and  $E[X^2]$ . Show that  $E[X^2] / E[X] > E[X]$ , confirming the Inspection Paradox.
- b) In stationarity, the probability that your arrival lands in a SHORT gap (length 2) is proportional to the gap length times its probability:  
$$P(\text{land in short gap}) = P(X=2) \cdot 2 / E[X]$$
$$P(\text{land in long gap}) = P(X=8) \cdot 8 / E[X]$$
Compute these probabilities. Notice that even though short gaps are more frequent (prob 0.6), you are more likely to land in a long gap.
- c) Given that you land in a gap of length  $L$ , you arrive at a  $\text{Uniform}[0, L]$  point within that gap (by symmetry, your arrival is equally likely at any position within the interval). Compute  $E[B \mid \text{short gap}] = E[U \mid U \sim \text{Uniform}[0, 2]]$  and  $E[B \mid \text{long gap}] = E[U \mid U \sim \text{Uniform}[0, 8]]$ , where  $B$  is the residual time until the next bus. Then use the law of total expectation to find  $E[B]$  and verify it equals  $E[X^2] / (2 \cdot E[X])$ .